

12.3-Support_Vector_Regression

August 7, 2025

1 Support Vector Regression Implementation

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import warnings
warnings.filterwarnings('ignore')
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
[2]: ## Dataset

df = sns.load_dataset('tips')
df.head()
```

```
[2]:   total_bill  tip  sex smoker  day  time  size
0      16.99  1.01 Female    No  Sun  Dinner     2
1      10.34  1.66   Male    No  Sun  Dinner     3
2      21.01  3.50   Male    No  Sun  Dinner     3
3      23.68  3.31   Male    No  Sun  Dinner     2
4      24.59  3.61 Female    No  Sun  Dinner     4
```

```
[3]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype
---  -
0   total_bill    244 non-null   float64
1   tip           244 non-null   float64
2   sex           244 non-null   category
3   smoker        244 non-null   category
4   day           244 non-null   category
5   time          244 non-null   category
```

```
6    size      244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

```
[4]: for i in df.columns:
      if df[i].dtype=='category':
          print("=="*20)
          print(df[i].value_counts())
```

```
=====
sex
Male      157
Female     87
Name: count, dtype: int64
=====
smoker
No        151
Yes        93
Name: count, dtype: int64
=====
day
Sat        87
Sun        76
Thur       62
Fri        19
Name: count, dtype: int64
=====
time
Dinner     176
Lunch       68
Name: count, dtype: int64
```

```
[5]: X = df.drop('total_bill', axis=1)
      y = df['total_bill']
```

```
[6]: X.head()
```

```
[6]:   tip  sex smoker  day  time  size
0  1.01 Female    No  Sun  Dinner    2
1  1.66  Male    No  Sun  Dinner    3
2  3.50  Male    No  Sun  Dinner    3
3  3.31  Male    No  Sun  Dinner    2
4  3.61 Female    No  Sun  Dinner    4
```

```
[7]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25,
    random_state=10)
```

```
[8]: X_train.head()
```

```
[8]:
```

	tip	sex	smoker	day	time	size
58	1.76	Male	Yes	Sat	Dinner	2
1	1.66	Male	No	Sun	Dinner	3
2	3.50	Male	No	Sun	Dinner	3
68	2.01	Male	No	Sat	Dinner	2
184	3.00	Male	Yes	Sun	Dinner	2

1.1 Feature Encoding (Label Encoding and OneHot Encoding)

```
[9]: from sklearn.preprocessing import LabelEncoder, OneHotEncoder
le1 = LabelEncoder()
le2 = LabelEncoder()
le3 = LabelEncoder()
```

```
[10]: X_train['sex'] = le1.fit_transform(X_train['sex'])
X_train['smoker'] = le2.fit_transform(X_train['smoker'])
X_train['time'] = le3.fit_transform(X_train['time'])
```

```
[11]: X_train.head()
```

```
[11]:
```

	tip	sex	smoker	day	time	size
58	1.76	1	1	Sat	0	2
1	1.66	1	0	Sun	0	3
2	3.50	1	0	Sun	0	3
68	2.01	1	0	Sat	0	2
184	3.00	1	1	Sun	0	2

```
[12]: X_test['sex'] = le1.transform(X_test['sex'])
X_test['smoker'] = le2.transform(X_test['smoker'])
X_test['time'] = le3.transform(X_test['time'])
```

```
[13]: X_test.head()
```

```
[13]:
```

	tip	sex	smoker	day	time	size
162	2.00	0	0	Sun	0	3
60	3.21	1	1	Sat	0	2
61	2.00	1	1	Sat	0	2
63	3.76	1	1	Sat	0	4
69	2.09	1	1	Sat	0	2

1.1.1 OneHot Encoding (ColumnTransformer)

```
[14]: from sklearn.compose import ColumnTransformer
ct = ColumnTransformer(transformers=[('onehot', OneHotEncoder(drop='first'),
↳ [3])],
remainder='passthrough')
```

```
[15]: X_train = ct.fit_transform(X_train)
```

```
[16]: X_test = ct.transform(X_test)
```

```
[17]: svr = SVR()  
svr.fit(X_train, y_train)
```

```
[17]: SVR()
```

```
[18]: y_pred = svr.predict(X_test)
```

```
[19]: print(f"r2_score:", r2_score(y_test, y_pred))  
print(f"mean_absolute_error:", mean_absolute_error(y_test, y_pred))  
print(f"mean_squared_error:", mean_squared_error(y_test, y_pred))
```

```
r2_score: 0.46028114561159283  
mean_absolute_error: 4.1486423210190235  
mean_squared_error: 45.692136151683385
```

1.2 Hyperparameter Tuning

```
[20]: from sklearn.model_selection import GridSearchCV
```

```
params = {  
    'C': [0.1, 1, 10, 100, 1000],  
    'gamma': [1, 0.1, 0.01, 0.001, 0.0001],  
    'kernel': ['rbf']  
}
```

```
[21]: grid = GridSearchCV(estimator=SVR(), param_grid=params, refit=True, verbose=3,  
    ↪cv = 5)
```

```
[22]: grid.fit(X_train, y_train)
```

Fitting 5 folds for each of 25 candidates, totalling 125 fits

```
[CV 1/5] END ...C=0.1, gamma=1, kernel=rbf;; score=-0.067 total time= 0.0s  
[CV 2/5] END ...C=0.1, gamma=1, kernel=rbf;; score=-0.058 total time= 0.0s  
[CV 3/5] END ...C=0.1, gamma=1, kernel=rbf;; score=-0.145 total time= 0.0s  
[CV 4/5] END ...C=0.1, gamma=1, kernel=rbf;; score=0.025 total time= 0.0s  
[CV 5/5] END ...C=0.1, gamma=1, kernel=rbf;; score=-0.089 total time= 0.0s  
[CV 1/5] END ...C=0.1, gamma=0.1, kernel=rbf;; score=0.013 total time= 0.0s  
[CV 2/5] END ...C=0.1, gamma=0.1, kernel=rbf;; score=0.021 total time= 0.0s  
[CV 3/5] END ...C=0.1, gamma=0.1, kernel=rbf;; score=-0.010 total time= 0.0s  
[CV 4/5] END ...C=0.1, gamma=0.1, kernel=rbf;; score=0.124 total time= 0.0s  
[CV 5/5] END ...C=0.1, gamma=0.1, kernel=rbf;; score=0.050 total time= 0.0s  
[CV 1/5] END ...C=0.1, gamma=0.01, kernel=rbf;; score=-0.053 total time= 0.0s  
[CV 2/5] END ...C=0.1, gamma=0.01, kernel=rbf;; score=-0.028 total time= 0.0s  
[CV 3/5] END ...C=0.1, gamma=0.01, kernel=rbf;; score=-0.108 total time= 0.0s  
[CV 4/5] END ...C=0.1, gamma=0.01, kernel=rbf;; score=0.040 total time= 0.0s
```

```

[CV 5/5] END ...C=0.1, gamma=0.01, kernel=rbf;; score=-0.058 total time= 0.0s
[CV 1/5] END ...C=0.1, gamma=0.001, kernel=rbf;; score=-0.080 total time= 0.0s
[CV 2/5] END ...C=0.1, gamma=0.001, kernel=rbf;; score=-0.068 total time= 0.0s
[CV 3/5] END ...C=0.1, gamma=0.001, kernel=rbf;; score=-0.167 total time= 0.0s
[CV 4/5] END ...C=0.1, gamma=0.001, kernel=rbf;; score=-0.006 total time= 0.0s
[CV 5/5] END ...C=0.1, gamma=0.001, kernel=rbf;; score=-0.105 total time= 0.0s
[CV 1/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=-0.083 total time= 0.0s
[CV 2/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=-0.073 total time= 0.0s
[CV 3/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=-0.173 total time= 0.0s
[CV 4/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=-0.013 total time= 0.0s
[CV 5/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=-0.110 total time= 0.0s
[CV 1/5] END ...C=1, gamma=1, kernel=rbf;; score=0.018 total time= 0.0s
[CV 2/5] END ...C=1, gamma=1, kernel=rbf;; score=0.051 total time= 0.0s
[CV 3/5] END ...C=1, gamma=1, kernel=rbf;; score=0.080 total time= 0.0s
[CV 4/5] END ...C=1, gamma=1, kernel=rbf;; score=0.166 total time= 0.0s
[CV 5/5] END ...C=1, gamma=1, kernel=rbf;; score=0.081 total time= 0.0s
[CV 1/5] END ...C=1, gamma=0.1, kernel=rbf;; score=0.189 total time= 0.0s
[CV 2/5] END ...C=1, gamma=0.1, kernel=rbf;; score=0.254 total time= 0.0s
[CV 3/5] END ...C=1, gamma=0.1, kernel=rbf;; score=0.533 total time= 0.0s
[CV 4/5] END ...C=1, gamma=0.1, kernel=rbf;; score=0.347 total time= 0.0s
[CV 5/5] END ...C=1, gamma=0.1, kernel=rbf;; score=0.503 total time= 0.0s
[CV 1/5] END ...C=1, gamma=0.01, kernel=rbf;; score=0.121 total time= 0.0s
[CV 2/5] END ...C=1, gamma=0.01, kernel=rbf;; score=0.221 total time= 0.0s
[CV 3/5] END ...C=1, gamma=0.01, kernel=rbf;; score=0.269 total time= 0.0s
[CV 4/5] END ...C=1, gamma=0.01, kernel=rbf;; score=0.305 total time= 0.0s
[CV 5/5] END ...C=1, gamma=0.01, kernel=rbf;; score=0.252 total time= 0.0s
[CV 1/5] END ...C=1, gamma=0.001, kernel=rbf;; score=-0.049 total time= 0.0s
[CV 2/5] END ...C=1, gamma=0.001, kernel=rbf;; score=-0.014 total time= 0.0s
[CV 3/5] END ...C=1, gamma=0.001, kernel=rbf;; score=-0.096 total time= 0.0s
[CV 4/5] END ...C=1, gamma=0.001, kernel=rbf;; score=0.050 total time= 0.0s
[CV 5/5] END ...C=1, gamma=0.001, kernel=rbf;; score=-0.051 total time= 0.0s
[CV 1/5] END ...C=1, gamma=0.0001, kernel=rbf;; score=-0.080 total time= 0.0s
[CV 2/5] END ...C=1, gamma=0.0001, kernel=rbf;; score=-0.068 total time= 0.0s
[CV 3/5] END ...C=1, gamma=0.0001, kernel=rbf;; score=-0.167 total time= 0.0s
[CV 4/5] END ...C=1, gamma=0.0001, kernel=rbf;; score=-0.006 total time= 0.0s
[CV 5/5] END ...C=1, gamma=0.0001, kernel=rbf;; score=-0.104 total time= 0.0s
[CV 1/5] END ...C=10, gamma=1, kernel=rbf;; score=0.166 total time= 0.0s
[CV 2/5] END ...C=10, gamma=1, kernel=rbf;; score=0.231 total time= 0.0s
[CV 3/5] END ...C=10, gamma=1, kernel=rbf;; score=0.468 total time= 0.0s
[CV 4/5] END ...C=10, gamma=1, kernel=rbf;; score=0.212 total time= 0.0s
[CV 5/5] END ...C=10, gamma=1, kernel=rbf;; score=0.377 total time= 0.0s
[CV 1/5] END ...C=10, gamma=0.1, kernel=rbf;; score=0.156 total time= 0.0s
[CV 2/5] END ...C=10, gamma=0.1, kernel=rbf;; score=0.433 total time= 0.0s
[CV 3/5] END ...C=10, gamma=0.1, kernel=rbf;; score=0.706 total time= 0.0s
[CV 4/5] END ...C=10, gamma=0.1, kernel=rbf;; score=0.395 total time= 0.0s
[CV 5/5] END ...C=10, gamma=0.1, kernel=rbf;; score=0.541 total time= 0.0s
[CV 1/5] END ...C=10, gamma=0.01, kernel=rbf;; score=0.204 total time= 0.0s
[CV 2/5] END ...C=10, gamma=0.01, kernel=rbf;; score=0.474 total time= 0.0s

```

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[CV 3/5] END ...C=10, gamma=0.01, kernel=rbf;; score=0.701 total time= 0.0s
[CV 4/5] END ...C=10, gamma=0.01, kernel=rbf;; score=0.432 total time= 0.0s
[CV 5/5] END ...C=10, gamma=0.01, kernel=rbf;; score=0.592 total time= 0.0s
[CV 1/5] END ...C=10, gamma=0.001, kernel=rbf;; score=0.144 total time= 0.0s
[CV 2/5] END ...C=10, gamma=0.001, kernel=rbf;; score=0.277 total time= 0.0s
[CV 3/5] END ...C=10, gamma=0.001, kernel=rbf;; score=0.338 total time= 0.0s
[CV 4/5] END ...C=10, gamma=0.001, kernel=rbf;; score=0.342 total time= 0.0s
[CV 5/5] END ...C=10, gamma=0.001, kernel=rbf;; score=0.283 total time= 0.0s
[CV 1/5] END ...C=10, gamma=0.0001, kernel=rbf;; score=-0.049 total time= 0.0s
[CV 2/5] END ...C=10, gamma=0.0001, kernel=rbf;; score=-0.013 total time= 0.0s
[CV 3/5] END ...C=10, gamma=0.0001, kernel=rbf;; score=-0.094 total time= 0.0s
[CV 4/5] END ...C=10, gamma=0.0001, kernel=rbf;; score=0.051 total time= 0.0s
[CV 5/5] END ...C=10, gamma=0.0001, kernel=rbf;; score=-0.050 total time= 0.0s
[CV 1/5] END ...C=100, gamma=1, kernel=rbf;; score=-0.040 total time= 0.0s
[CV 2/5] END ...C=100, gamma=1, kernel=rbf;; score=0.190 total time= 0.0s
[CV 3/5] END ...C=100, gamma=1, kernel=rbf;; score=0.429 total time= 0.0s
[CV 4/5] END ...C=100, gamma=1, kernel=rbf;; score=-0.095 total time= 0.0s
[CV 5/5] END ...C=100, gamma=1, kernel=rbf;; score=0.242 total time= 0.0s
[CV 1/5] END ...C=100, gamma=0.1, kernel=rbf;; score=0.147 total time= 0.0s
[CV 2/5] END ...C=100, gamma=0.1, kernel=rbf;; score=0.537 total time= 0.0s
[CV 3/5] END ...C=100, gamma=0.1, kernel=rbf;; score=0.710 total time= 0.0s
[CV 4/5] END ...C=100, gamma=0.1, kernel=rbf;; score=0.499 total time= 0.0s
[CV 5/5] END ...C=100, gamma=0.1, kernel=rbf;; score=0.354 total time= 0.0s
[CV 1/5] END ...C=100, gamma=0.01, kernel=rbf;; score=0.154 total time= 0.0s
[CV 2/5] END ...C=100, gamma=0.01, kernel=rbf;; score=0.515 total time= 0.1s
[CV 3/5] END ...C=100, gamma=0.01, kernel=rbf;; score=0.727 total time= 0.1s
[CV 4/5] END ...C=100, gamma=0.01, kernel=rbf;; score=0.368 total time= 0.5s
[CV 5/5] END ...C=100, gamma=0.01, kernel=rbf;; score=0.581 total time= 0.0s
[CV 1/5] END ...C=100, gamma=0.001, kernel=rbf;; score=0.215 total time= 0.0s
[CV 2/5] END ...C=100, gamma=0.001, kernel=rbf;; score=0.502 total time= 0.0s
[CV 3/5] END ...C=100, gamma=0.001, kernel=rbf;; score=0.717 total time= 0.0s
[CV 4/5] END ...C=100, gamma=0.001, kernel=rbf;; score=0.420 total time= 0.0s
[CV 5/5] END ...C=100, gamma=0.001, kernel=rbf;; score=0.601 total time= 0.0s
[CV 1/5] END ...C=100, gamma=0.0001, kernel=rbf;; score=0.146 total time= 0.0s
[CV 2/5] END ...C=100, gamma=0.0001, kernel=rbf;; score=0.284 total time= 0.0s
[CV 3/5] END ...C=100, gamma=0.0001, kernel=rbf;; score=0.346 total time= 0.0s
[CV 4/5] END ...C=100, gamma=0.0001, kernel=rbf;; score=0.346 total time= 0.0s
[CV 5/5] END ...C=100, gamma=0.0001, kernel=rbf;; score=0.288 total time= 0.0s
[CV 1/5] END ...C=1000, gamma=1, kernel=rbf;; score=-0.908 total time= 0.0s
[CV 2/5] END ...C=1000, gamma=1, kernel=rbf;; score=0.139 total time= 0.0s
[CV 3/5] END ...C=1000, gamma=1, kernel=rbf;; score=0.040 total time= 0.0s
[CV 4/5] END ...C=1000, gamma=1, kernel=rbf;; score=-1.755 total time= 0.0s
[CV 5/5] END ...C=1000, gamma=1, kernel=rbf;; score=0.056 total time= 0.0s
[CV 1/5] END ...C=1000, gamma=0.1, kernel=rbf;; score=0.042 total time= 0.0s
[CV 2/5] END ...C=1000, gamma=0.1, kernel=rbf;; score=0.493 total time= 0.0s
[CV 3/5] END ...C=1000, gamma=0.1, kernel=rbf;; score=0.114 total time= 0.0s
[CV 4/5] END ...C=1000, gamma=0.1, kernel=rbf;; score=0.048 total time= 0.0s
[CV 5/5] END ...C=1000, gamma=0.1, kernel=rbf;; score=0.284 total time= 0.0s

```

```

[CV 1/5] END ..C=1000, gamma=0.01, kernel=rbf;, score=0.120 total time= 0.0s
[CV 2/5] END ..C=1000, gamma=0.01, kernel=rbf;, score=0.572 total time= 0.0s
[CV 3/5] END ..C=1000, gamma=0.01, kernel=rbf;, score=0.752 total time= 0.0s
[CV 4/5] END ..C=1000, gamma=0.01, kernel=rbf;, score=0.425 total time= 0.0s
[CV 5/5] END ..C=1000, gamma=0.01, kernel=rbf;, score=0.517 total time= 0.0s
[CV 1/5] END ..C=1000, gamma=0.001, kernel=rbf;, score=0.217 total time= 0.0s
[CV 2/5] END ..C=1000, gamma=0.001, kernel=rbf;, score=0.518 total time= 0.0s
[CV 3/5] END ..C=1000, gamma=0.001, kernel=rbf;, score=0.716 total time= 0.0s
[CV 4/5] END ..C=1000, gamma=0.001, kernel=rbf;, score=0.347 total time= 0.0s
[CV 5/5] END ..C=1000, gamma=0.001, kernel=rbf;, score=0.635 total time= 0.0s
[CV 1/5] END ..C=1000, gamma=0.0001, kernel=rbf;, score=0.216 total time= 0.0s
[CV 2/5] END ..C=1000, gamma=0.0001, kernel=rbf;, score=0.505 total time= 0.0s
[CV 3/5] END ..C=1000, gamma=0.0001, kernel=rbf;, score=0.718 total time= 0.0s
[CV 4/5] END ..C=1000, gamma=0.0001, kernel=rbf;, score=0.420 total time= 0.0s
[CV 5/5] END ..C=1000, gamma=0.0001, kernel=rbf;, score=0.602 total time= 0.0s

```

```

[22]: GridSearchCV(cv=5, estimator=SVR(),
                  param_grid={'C': [0.1, 1, 10, 100, 1000],
                              'gamma': [1, 0.1, 0.01, 0.001, 0.0001],
                              'kernel': ['rbf']},
                  verbose=3)

```

```

[23]: grid.best_params_

```

```

[23]: {'C': 1000, 'gamma': 0.0001, 'kernel': 'rbf'}

```

```

[24]: grid.best_score_

```

```

[24]: 0.492086225466603

```

```

[25]: grid_prediction = grid.predict(X_test)

```

```

[26]: print(f'r2_score:', r2_score(y_test, grid_prediction))
      print(f"mean_absolute_error:", mean_absolute_error(y_test, grid_prediction))
      print(f"mean_squared_error:", mean_squared_error(y_test, grid_prediction))

```

```

r2_score: 0.5081618245078687
mean_absolute_error: 3.8685147526100234
mean_squared_error: 41.63859886764171

```

```

[ ]:

```

```

[ ]:

```

```

[ ]:

```